

Intermediate Elective

Zelda Reif

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Set Up

Packages

```
# general use
library(tidyverse)
library(snakecase)
library(janitor)

# file organization
library(here)

# reading .xlsx files
library(readxl)

# data visualization
library(naniar)
library(patchwork)
library(NatParksPalettes)
library(gghighlight)
library(paletteer)
library(ggplot2)
```

Data

```
# taxonomic info
taxon_list <- read_csv(here("data","taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site info
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")
```

Problems

1. Explain your inspiration

I was inspired by Tobias Stalder's "Hiking Locations in Washington" because the figure's characteristics would allow me to show average abundance/L data for a specific aquatic invertebrate species, separated by NCOS location, while also highlighting a mean average abundance/L value for each site.

This figure makes sense for the aquatic invertebrate data set because it allows for a clear comparison of average abundance/L between different sites, just as the original figure is separated by hiking locations.

The plot will show average abundance/L of ostracods across NCOS locations, the x axis will be NCOS location and the y axis will be average abundance/L. Each jitter point will represent a different average abundance/L measurement, and the mean average abundance/L for each location will be highlighted with a different color to stand out.

2.

Sketch what your figure would look like following your inspiration.

Insert that sketch (as a photo, screenshot, or otherwise) into your document.

3. Code your figure

Cleaning

```
taxon_list_clean <- taxon_list |>
  # convert taxon names to snakecase
  mutate(verbatim_name = to_any_case(verbatim_name,
    case = "snake"))

# creating new object
ncos_sites <- sites |>
# filtering to only include ncos quarterly sampling sites
filter(sector == "NCOS" & sampling_frequency == "Quarterly")

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

water_quality_clean <- water_quality |>
  clean_names() |>
  semi_join(ncos_sites, by = "site") |>
  unite("date_site", date, site, remove = FALSE) |>
  group_by(date, site, date_site) |>
  summarize(med_ph = median(p_h, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),
    med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  ungroup() |>
  filter(date_site != "2022-08-12_NVBR")

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
    ostracod,
```

```

    copepod,
    hemiptera_corixidae_boatman,
    cladocera,
    nematode,
    diptera_ceratopogonidae,
    annelida_oligochaete,
    diptera_chironomid,
    annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
              names_to = "taxon",
              values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5
# liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(
  select(water_quality_clean,
        date_site,
        med_ph,
        med_sal,
        med_do),
        by = "date_site") |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek")) |>

```

```

# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
= TRUE),
site_full = fct_rev(site_full))

# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
# filter to only include copepods
filter(taxon == "copepod") |>
# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))

```

Wrangling

```

# create new object from aq_ins_clean
ost_abundance <- aq_ins_clean |>
# filter to only ostracods
filter(taxon == "ostracod") |>
# remove missing abundance/L values
filter(!is.na(ave_lit)) |>
# group by site
group_by(site_full) |>
# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1)) |>
# ungroup data frame
ungroup()

```

Plotting

```

y_max <- max(ost_abundance$log_ave_lit, na.rm = TRUE)
ring_log <- pretty(c(0, y_max), n = 4)

ostracods_plot <- ggplot(
  data = ost_abundance,
  mapping = aes(
    x = site_full,
    y = log_ave_lit

```

```

)
) +

geom_hline(
  yintercept = ring_log,
  color = "lightgray"
) +

geom_vline(
  xintercept = 1:length(unique(ost_abundance$site_full)),
  linetype = "dashed",
  color = "gray30"
) +

geom_jitter(
  height = 0,
  width = 0.1,
  shape = 21
) +

stat_summary(
  geom = "point",
  fun = "median",
  size = 2,
  color = "lightblue"
) +

scale_x_discrete(labels = \(x) str_wrap(x, 15)) +

#scale y axis so bars don't start in the center
scale_y_continuous(limits = c(-1500, 3500),
  expand = c(0,0),
  breaks = c(0, 1000, 2000, 3000))+

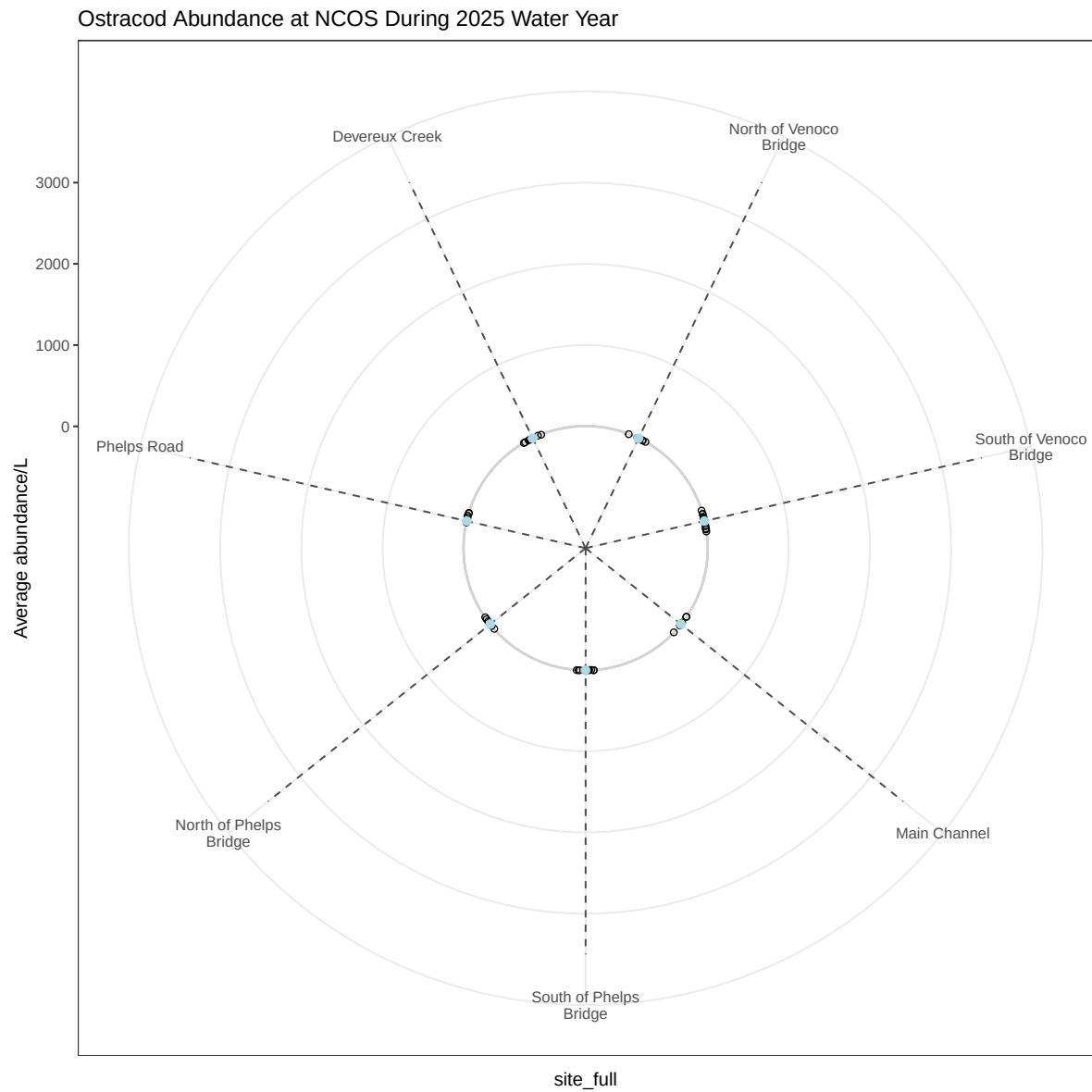
labs(y = "Average abundance/L",
  title = "Ostracod Abundance at NCOS During 2025 Water Year"
) +

coord_polar() +

theme_bw()

```

ostracods_plot



```
ostracods_plot <- ggplot(  
  data = ost_abundance,  
  mapping = aes(  
    x = site_full,  
    y = log_ave_lit
```

```

)
) +

geom_hline(
  yintercept = ring_log,
  color = "lightgray"
) +

geom_vline(
  xintercept = 1:length(unique(ost_abundance$site_full)),
  linetype = "dashed",
  color = "gray30"
) +

geom_jitter(
  height = 0,
  width = 0.1,
  shape = 21
) +

stat_summary(
  geom = "point",
  fun = "median",
  size = 2,
  color = "lightblue"
) +

scale_x_discrete(labels = \(x) str_wrap(x, 15)) +

labs(
  x = "Site",
  y = "Average abundance/L",
  title = "Ostracod Abundance at NCOS During 2025 Water Year"
) +

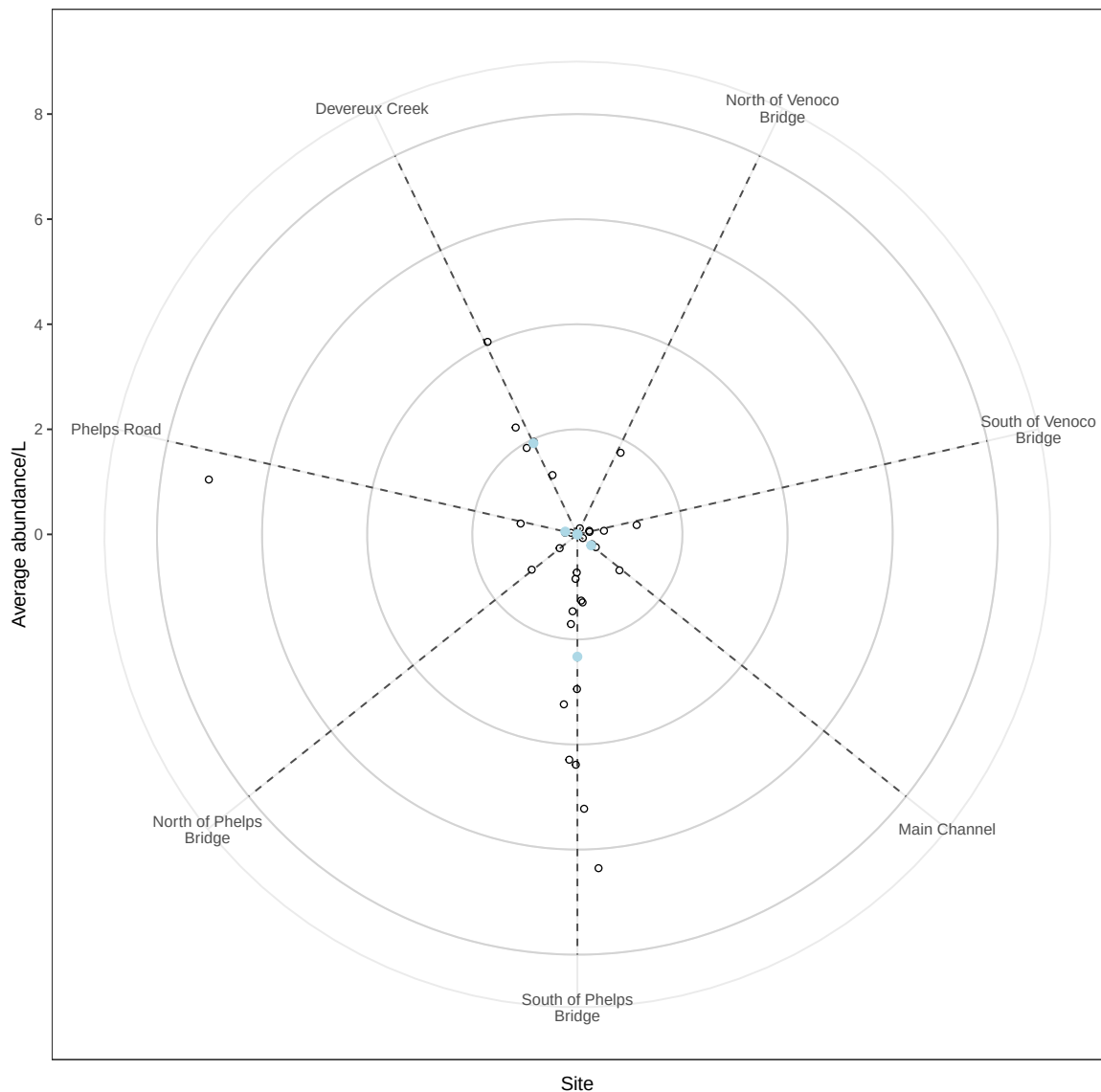
coord_polar() +

theme_bw()

ostracods_plot

```


Ostracod Abundance at NCOS During 2025 Water Year



4. Describe your process

In 1-3 sentences each, describe:

- your process of using someone else's code to create a visualization
- challenges you encountered as you made your visualization
- how your visualization differs from visualizations you made in class, for assignments, or for a previous elective